

Control a plant watering system using the internet

In this DIY, we'll make a web-enabled plant watering system. This will allow you to water the plants in your home, building or even colony while sitting in another part of the world. For this purpose, we'll create a web server that will listen to a request for switching the motor ON and OFF.

Hardware needed:

- ◆ Raspberry Pi
- ◆ Breadboard
- ◆ Water pump / Solenoid valve
- ◆ Solid State Relay
- ◆ N3904 Transistor
- ◆ 4.7 k ohm Resistor

The above mentioned list includes all the electronic parts of the complete setup. You'll also need the required piped connections for distributing water.

Let's get started by adding logic to the Raspberry Pi. First, install the Python GPIO wrapper by using the following command in the terminal

```
• sudo apt-get install
  rpi.gpio
```

Install the Python server Twisted, using the command:

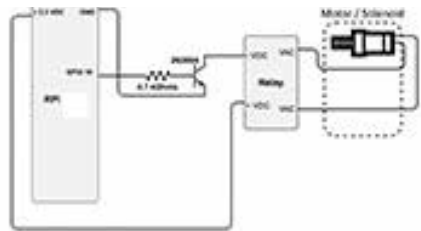
- ◆ **sudo apt-get install python-twisted**

Connect the circuit as shown in the figure. Now, in the terminal enter the following command to create a Python script that will control the motor or solenoid:

```
• nano basic_switch.py
```



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Connection for plant watering system.